RiskRadar

From raw news to risk signals — an AI/NLP Risk Engine that rebalances an index and stress-tests a bank portfolio, live.

`2+ sources · news + social` `3 signals · sentiment / event / impact` `BOTH modules · rebalancing + stress testing`

`0.3 ms per signal · fully offline`

Phase III — Case Study Competition · Individual submission

Aare Renu Shri · add roll ID here

Markets move on news. Risk systems only eat numbers.

The case study asks for an engine that converts real-time unstructured text into structured risk signals — then at least one module that acts on them. I built the core **and both** modules.

CORE (REQUIRED)

AI/NLP Risk Engine

Ingests **financial news + social posts** (2 source types; live NewsAPI/GDELT/Reddit connectors optional, fail-soft). Every item → **sentiment (–1…+1)**, **event class (12 types)**, **impact (1–10)** — each carrying its own evidence.

MODULE A (BUILT)

Index Rebalancing

Subscribes to sentiment; tilts a 20-stock mock S&P-100 index with w×(1+0.25·s), 2–12% band, full audit trail — weights drift live on the dashboard.

MODULE B (BUILT)

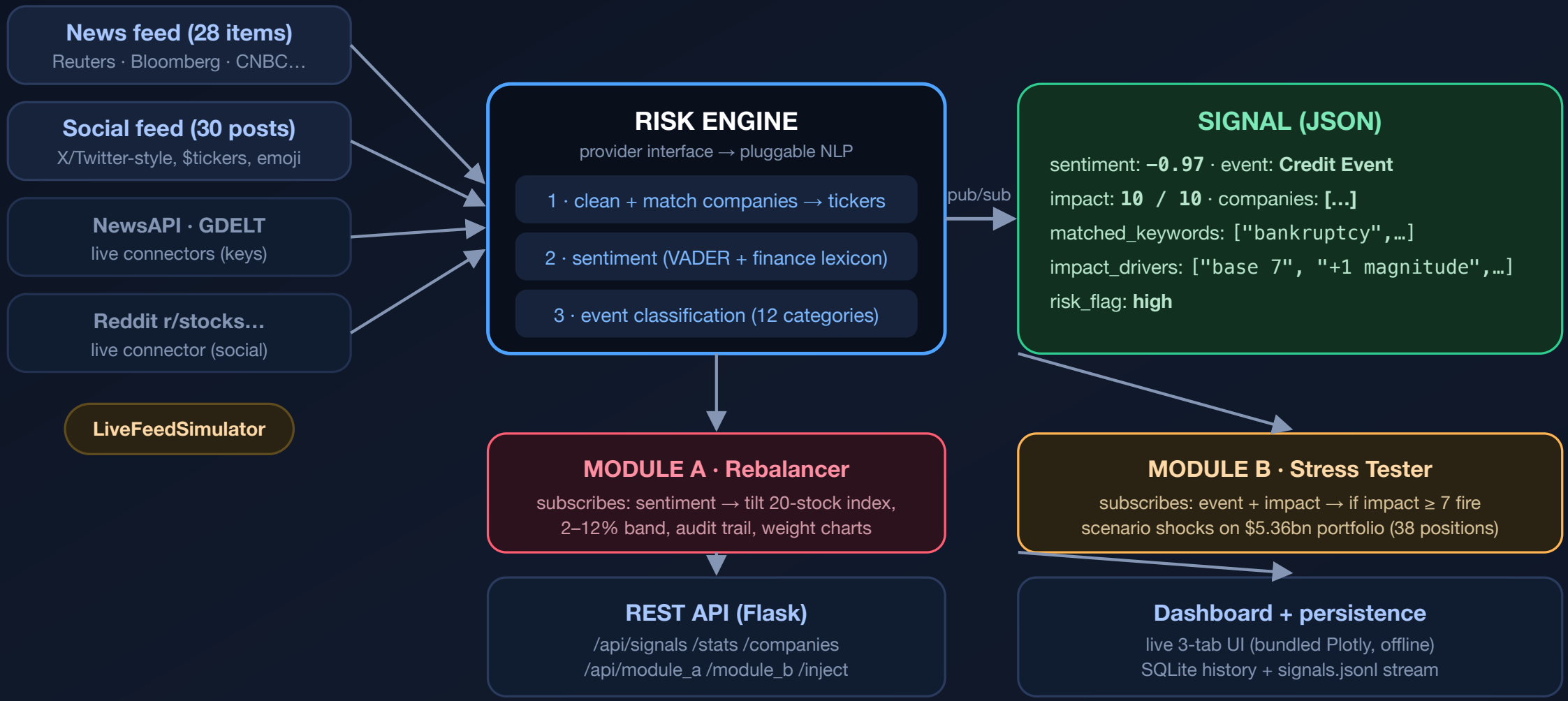
Stress Testing

Subscribes to event + impact; any **impact ≥ 7** event fires a scenario shock (e.g. rates +200bp, equity −10%) on a **$5.36bn** wholesale portfolio — 38 positions, 9 asset classes — with before/after P&L.

> "One engine, three explainable numbers, two consumers — tactical rebalancing and strategic stress testing — running live, offline, on a laptop."

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ARCHITECTURE · ONE PROCESS, ONE DATA-FLOW DIRECTION



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THE CORE · EVERY TEXT → THREE NUMBERS + EVIDENCE

sentiment –1…+1 · VADER + 150 finance terms

event · 12 classes, priority rules

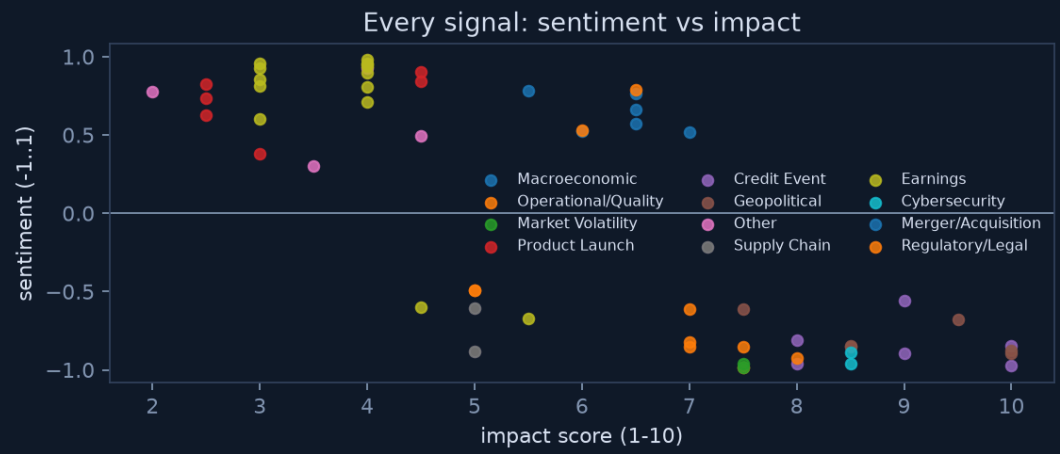
impact 1–10 · transparent formula

```
IN : "OPEC+ surprise production cut escalates
       trade tensions with Washington"
OUT: { "sentiment": -0.84,
       "event_type": "Geopolitical",
       "impact": 8.5,  "risk_flag": "high",
       "matched_keywords": ["opec","escalating",…],   // why this
class
       "impact_drivers": ["base 6.5 (Geo)",
                          "+1 urgency"," +1 negative"] }          // why
8.5
```

93% of 58 corpus items got a meaningful class; 54/58 · 0.3 ms/signal · ~3,300/sec on one core.

Every signal carries its evidence — keywords + drivers — because a risk number that can't answer "why?" is worthless.

```
demo killer move: type any unseen headline in the dashboard →
verdict in 0.3 ms
```



every signal: sentiment × impact, colored by event class — the red zone is where Module B wakes up

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MODULE A · TACTICAL — SENTIMENT → INDEX WEIGHTS, LIVE

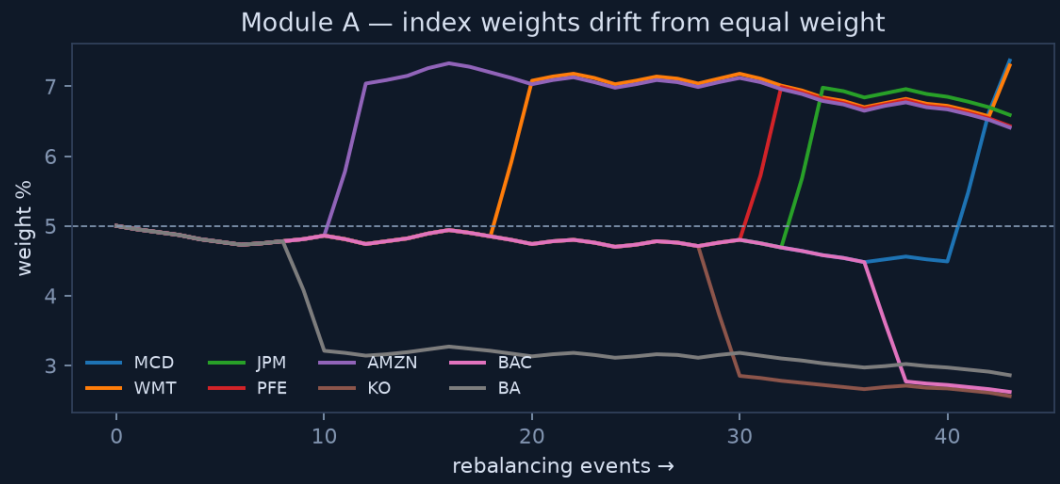
```
w_new = w_old × (1 + 0.25 × sentiment)
w     = clamp(w, 2%, 12%)      // concentration limits
w    /= Σ w                    // renormalize to 100%
```

- **20 stocks** (S&P 100 subset), start equal-weighted at 5%
- **45 weight changes** logged in the demo run, each with the signal that caused it — full audit trail
- Positive sentiment earns share; negative sheds it. Compounds across the stream — that's the drift you see below.

WINNERS
MCD 7.4%
WMT 7.3%
JPM 6.6%

LOSERS
KO 2.6%
BAC 2.6%
BA 2.9%

WHY
recall −0.98 · subprime warnings · supply-chain hit



weights drifting from the 5% equal-weight line as sentiment arrives

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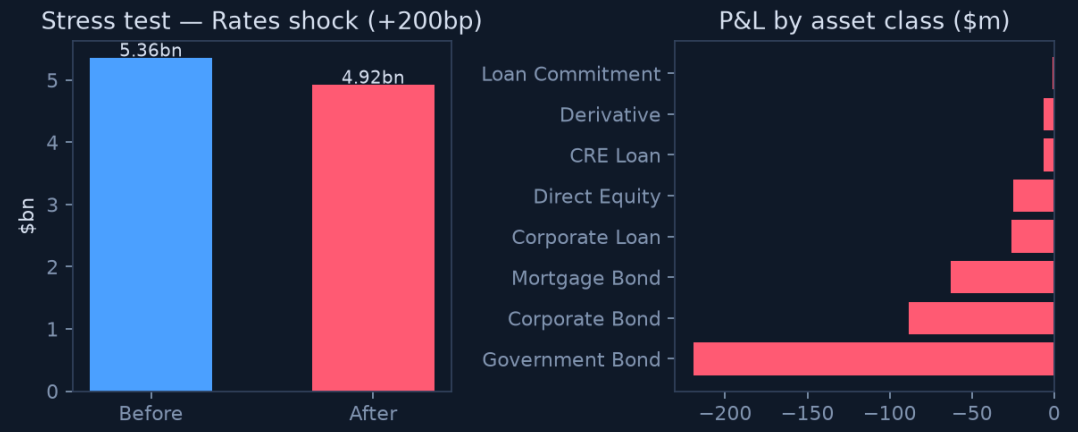
MODULE B · STRATEGIC — HIGH-IMPACT EVENTS FIRE STRESS SCENARIOS

- **$5.36bn synthetic wholesale portfolio** — 38 positions, 9 asset classes: corporate loans $1.6bn, CRE, govt bonds, IG/HY credit, MBS, derivatives (swaps, CDS), commitments, equity.
- **Trigger: impact ≥ 7** (the case study's own threshold) → event-typed shock:
  `rates +200bp` `spreads +250bp` `equity −10%`
- First-order, fully traceable math: bonds lose `notional × duration × Δrate` — the 30Y Treasury alone bleeds **$92.5m** in the rates shock.

RATES SHOCK
−8.1%
−$435m
Macroeconomic · impact 9

CREDIT SHOCK
−6.6%
−$355m
Chapter 11 · impact 10

GEO SHOCK
−4.0%
−$215m
OPEC cut · impact 8.5



before/after valuation + P&L by asset class — you can see exactly where the loss lands

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CHALLENGES · HONEST LIMITS · WHAT'S NEXT

HARDEST PROBLEMS WE SOLVED

- Unreliable live APIs → **fail-soft architecture** + live-feed simulator; demo can't break
- VADER didn't know finance → **150-term lexicon**; classification conflicts → priority rules
- "Impact 7" had no definition → **transparent formula**, drivers stored on every signal

HONEST LIMITATIONS

- Sentiment is lexicon-based (v1), not learned — **chosen for explainability**, not hidden power
- Impact bases are domain-judged, not back-calibrated on history
- Stress math is first-order — no VaR, correlations, netting

ONE MORE WEEK →

- **FinBERT** behind the provider interface (one-file swap)
- Backtest Module A on the case study's Kaggle tweet/price datasets
- Scenario-VaR layer for Module B · Kafka/WebSockets · auth + Docker

"One engine · three explainable numbers · two consumers — tactical rebalancing and strategic stress testing, running live, offline, on one laptop. Ask me anything — or throw any headline at it right now."

`python run.py` `dashboard → localhost:8000` `repo: RiskRadar/` `report: Project Report & Q&A PDF`